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United States Patent	12397611
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Kang; Su Yeon et al.

Apparatus and system for sharing air conditioning air for mobility vehicle

Abstract

In an exemplary embodiment of the present disclosure, a mobility vehicle and a tent shares air-conditioned air through a duct so that the air-conditioned air produced in the mobility vehicle circulates through the duct and a temperature in an internal space of the tent is adjusted. Furthermore, in an apparatus and system for sharing air-conditioned air for a mobility vehicle, energy efficiency is improved as a loss of the air-conditioned air is minimized at the time of transmitting the air-conditioned air from the mobility vehicle to the tent.

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Appl. No.: **17/869577**

Filed: **July 20, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230138139 A1	May. 04, 2023

Foreign Application Priority Data

KR	10-2021-0148779	Nov. 02, 2021

Publication Classification

Int. Cl.: **B60H1/00** (20060101); **B60P3/36** (20060101)

U.S. Cl.:

CPC **B60H1/00407** (20130101); **B60H1/00028** (20130101); **B60P3/36** (20130101); B60H2001/00085 (20130101)

Field of Classification Search

CPC: B60P (3/36); B60P (3/38); B60P (3/39); B60P (3/32); B60H (1/00407); B60H (1/00564)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) The present application claims priority to Korean Patent Application No. 10-2021-0148779, filed Nov. 2, 2021, the entire contents of which is incorporated herein for all purposes by this reference.

BACKGROUND OF THE PRESENT DISCLOSURE

Field of the Present Disclosure

(2) The present disclosure relates to an apparatus and system for sharing air-conditioned air for a mobility vehicle, which allow air-conditioned air produced by a mobility vehicle to be shared with a tent during outdoor camping so that a temperature in an internal space of the tent is adjusted by the air conditioning air, ensuring pleasant use of the tent.

Description of Related Art

(3) Recently, there have been developed mobility vehicles using electricity stored in batteries as well as mobility vehicles that travel using fossil fuel such as gasoline, diesel, gas, and the like.

(4) Furthermore, technologies are being developed so that the mobility vehicle, which operates using electricity, not only performs a traveling function, but also shares electricity with other devices.

(5) Therefore, a user may perform leisure activities by use of the mobility vehicle. To perform the leisure activities using the mobility vehicle, the user connects a separate camping vehicle or trailer

to the mobility vehicle or simply accommodates a tent in the mobility vehicle and then installs the tent outside the interior of the mobility vehicle.

(6) In the instant case, the camping vehicle or trailer may implement various functions including an air conditioning function, but utilization of the camping vehicle or trailer deteriorates because the camping vehicle or trailer has a large volume and is limited in moving.

(7) Therefore, the user tends to accommodate a tent in the mobility vehicle, moves to a location, install the tent at the location, and then enjoy the leisure.

(8) However, because the tent is provided separately from the mobility vehicle and has no separate air conditioning facility, it is difficult to ensure a pleasant internal of the tent.

(9) The information included in this Background of the present disclosure section is only for enhancement of understanding of the general background of the present disclosure and may not be taken as an acknowledgement or any form of suggestion that this information forms the prior art already known to a person skilled in the art.

BRIEF SUMMARY

(10) Various aspects of the present disclosure are directed to providing an apparatus and system for sharing air-conditioned air for a mobility vehicle, which allow air-conditioned air produced by a mobility vehicle to be shared with a tent during outdoor camping so that a temperature in an internal space of the tent is adjusted by the air conditioning air, ensuring pleasant use of the tent.

(11) Various aspects of the present disclosure are directed to providing an apparatus of sharing air-conditioned air for a mobility vehicle, the apparatus including: a mobility vehicle including an air conditioner configured to provide the air-conditioned air to an interior of the mobility vehicle; a shelter provided outside the interior of the mobility vehicle and including an internal space in the shelter; and a duct configured for extending to allow the interior of the mobility vehicle and the internal space of the shelter to fluidically communicate with each other, the duct being configured to allow the air-conditioned air provided by the air conditioner to be shared with the internal space of the shelter.

(12) The mobility vehicle may have an air intake port through which internal air of the mobility vehicle is introduced, and an air discharge port through which the air-conditioned air provided by the air conditioner is discharged, the duct may include an inlet duct and an outlet duct, the inlet duct may be connected to the air discharge port to allow the air-conditioned air to flow to the shelter, and the outlet duct may be connected to the air intake port to allow air in the shelter to circulate through the mobility vehicle.

(13) One end portion of the inlet duct may be detachably connected to the air discharge port of the mobility vehicle, the other end portion of the inlet duct may be penetratively connected to the shelter, one end portion of the outlet duct may be penetratively connected to the shelter, and the other end portion of the outlet duct may be detachably connected to the air intake port of the mobility vehicle.

(14) The inlet duct may be formed to surround the air discharge port and mounted to cover the air discharge port.

(15) When the air-conditioned air is provided through a front air conditioner, the inlet duct may be coupled to a defrosting discharge port among the air discharge ports.

(16) When the air-conditioned air is provided through a rear air conditioner, the inlet duct may be connected to the air discharge port connected to the rear air conditioner, and the outlet duct may be connected to an air inlet port connected to the rear air conditioner.

(17) The apparatus may further include a duct bracket detachably mounted on the mobility vehicle and configured to fix a position of the duct.

(18) The duct bracket may be detachably mounted on an opening/closing unit including a door glass or a roof of the mobility vehicle, and the duct may pass through the duct bracket to allow the interior of the mobility vehicle and the internal space of the shelter to fluidically communicate with each other through the duct.

(19) The duct bracket may be formed to match a shape of a portion of the door glass or the roof forming the opening/closing unit, and the duct bracket may be mounted by being pressed against the door glass or the roof when the opening/closing unit is closed.
(20) Various aspects of the present disclosure are directed to providing a system for sharing air-conditioned air for a mobility vehicle, the system including: a mobility vehicle including an air conditioner configured to provide the air-conditioned air to an interior of the mobility vehicle, the air conditioner including an air intake port through which internal air of the mobility vehicle is introduced, and an air discharge port through which the air-conditioned air provided by the air conditioner is discharged; a shelter provided outside the interior of the mobility vehicle and including an internal space in the shelter; a duct configured for extending to allow the interior of the mobility vehicle and the internal space of the shelter to fluidically communicate with each other, the duct being configured to allow the air-conditioned air provided by the air conditioner to be shared with the internal space of the shelter; and a control unit configured to control the mobility vehicle including the air conditioner, include a camping mode in addition to an air conditioning mode, and control the air conditioner according to the camping mode when the camping mode is selected.
(21) When the camping mode is selected, the control unit may perform a recirculation mode in which outside air is blocked out of the mobility vehicle and internal air circulates in the mobility vehicle.
(22) When the camping mode is selected, the control unit may blow the air-conditioned air at a maximum flow rate.
(23) The duct may include an inlet duct and an outlet duct, the inlet duct may be connected to the air discharge port to allow the air-conditioned air to flow to the shelter, the outlet duct may be connected to the air intake port to allow air in the shelter to circulate through the mobility vehicle, and when the camping mode is selected, the control unit may perform control to open the air discharge port connected to the inlet duct and close a remaining air discharge port.
(24) The control unit may receive information as to whether an occupant is present in the interior of the mobility vehicle, and when the occupant is present in the interior of the mobility vehicle, the control unit may perform control to open the air discharge port corresponding to a seat in which the occupant is accommodated.
(25) When the camping mode is selected, the control unit may check, from a user, whether the inlet duct and the outlet duct are mounted, and when the control unit concludes, from the user, that the inlet duct and the outlet duct are mounted, the control unit may perform control of the air conditioner according to the camping mode.
(26) The mobility vehicle may further include a duct bracket detachably mounted on an opening/closing unit including a door glass or a roof of the mobility vehicle, the duct bracket may be configured to fix the inlet duct and the outlet duct as the inlet duct and the outlet duct penetrate the duct bracket, and the control unit may check whether the duct bracket is mounted on the opening/closing unit when the camping mode is selected.
(27) When an opening amount of the opening/closing unit is at a predetermined level, the control unit may determine that the duct bracket is mounted, and when the control unit determines that the duct bracket is mounted, the control unit may perform control of the air conditioner according to the camping mode.
(28) When the camping mode is selected, the control unit may perform control of the air conditioner not to perform a function of removing moisture.
(29) When the camping mode is selected, the control unit may receive information in a state of charge (SOC) value of a battery of the mobility vehicle and check a charging station closest to a current position of the mobility vehicle or a minimum amount of electricity of the battery which is to be consumed while the mobility vehicle gets a preset charging station, and when the state of charge of the battery reaches the minimum amount of electricity of the battery, the control unit may

not operate the air conditioner.
(30) The apparatus and system for sharing air-conditioned air for the mobility vehicle structured as described above allow the mobility vehicle and the shelter to share the air-conditioned air through the duct so that the air-conditioned air produced in the mobility vehicle circulates through the duct and the temperature in the internal space of the shelter is adjusted.
(31) Furthermore, energy efficiency is improved as a loss of the air-conditioned air is minimized at the time of transmitting the air-conditioned air from the mobility vehicle to the shelter.
(32) The methods and apparatuses of the present disclosure have other features and advantages which will be apparent from or are set forth in more detail in the accompanying drawings, which are incorporated herein, and the following Detailed Description, which together serve to explain certain principles of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a view exemplarily illustrating an apparatus of sharing air-conditioned air for a mobility vehicle according to various exemplary embodiments of the present disclosure.
(2) FIG. **2** is a view exemplarily illustrating connection of a duct of the apparatus of sharing air-conditioned air for a mobility vehicle illustrated in FIG. **1**.
(3) FIG. **3** is a view exemplarily illustrating an apparatus of sharing air-conditioned air for a mobility vehicle according to various exemplary embodiments of the present disclosure.
(4) FIG. **4** is a view exemplarily illustrating a duct bracket according to an exemplary embodiment of the present disclosure.
(5) FIG. **5** is a configuration view exemplarily illustrating a system for sharing air-conditioned air for a mobility vehicle.
(6) FIG. **6** is a flowchart illustrating a method of controlling sharing air-conditioned air for a mobility vehicle.
(7) It may be understood that the appended drawings are not necessarily to scale, presenting a somewhat simplified representation of various features illustrative of the basic principles of the present disclosure. The specific design features of the present disclosure as included herein, including, for example, specific dimensions, orientations, locations, and shapes will be determined in part by the particularly intended application and use environment.
(8) In the figures, reference numbers refer to the same or equivalent parts of the present disclosure throughout the several figures of the drawing.
DETAILED DESCRIPTION
(9) Reference will now be made in detail to various embodiments of the present disclosure(s), examples of which are illustrated in the accompanying drawings and described below. While the present disclosure(s) will be described in conjunction with exemplary embodiments of the present disclosure, it will be understood that the present description is not intended to limit the present disclosure(s) to those exemplary embodiments of the present disclosure. On the other hand, the present disclosure(s) is/are intended to cover not only the exemplary embodiments of the present disclosure, but also various alternatives, modifications, equivalents and other embodiments, which may be included within the spirit and scope of the present disclosure as defined by the appended claims.
(10) Hereinafter, an apparatus and system for sharing air-conditioned air for a mobility vehicle according to various exemplary embodiments of the present disclosure will be described with reference to the accompanying drawings.
(11) FIG. **1** is a view exemplarily illustrating an apparatus of sharing air-conditioned air for a mobility vehicle according to various exemplary embodiments of the present disclosure, FIG. **2** is a

view exemplarily illustrating connection of a duct of the apparatus of sharing air-conditioned air for a mobility vehicle illustrated in FIG. **1**, FIG. **3** is a view exemplarily illustrating an apparatus of sharing air-conditioned air for a mobility vehicle according to various exemplary embodiments of the present disclosure, FIG. **4** is a view exemplarily illustrating a duct bracket according to an exemplary embodiment of the present disclosure, FIG. **5** is a configuration view exemplarily illustrating a system for sharing air-conditioned air for a mobility vehicle, and FIG. **6** is a flowchart illustrating a method of controlling sharing air-conditioned air for a mobility vehicle.

(12) As illustrated in FIG. **1**, an apparatus of sharing air-conditioned air for a mobility vehicle according to an exemplary embodiment of the present disclosure a mobility vehicle **100** including an air conditioner **110** configured to provide air-conditioned air to an interior of the mobility vehicle **100**; a tent **200** provided outside the interior of the mobility vehicle **100** and including an internal space; and a duct **300** extending to allow the interior of the mobility vehicle **100** and the internal space of the tent **200** to fluidically communicate with each other, the duct **300** being configured to allow the air-conditioned air provided by the air conditioner **110** to be shared with the internal space of the tent **200**.

(13) The mobility vehicle **100** may be operated by electric power of a battery and provide the air-conditioned air to the internal thereof to cool or heat the internal thereof. Therefore, the air conditioner **110** may produce cooling air by operating an electric compressor or produce heating air by operating a heat generator. The air conditioner **110** may provide the cooling air and the heating air and improve air conditioning efficiency by operating a heat pump.

(14) The tent **200** is provided outside the interior of the mobility vehicle **100**, provided separately from the mobility vehicle **100**, and has the internal space.

(15) The mobility vehicle **100** and the tent **200** are connected to fluidically communicate with each other by the duct **300**, and the air-conditioned air provided into the interior of the mobility vehicle **100** flows and circulates through the tent **200** so that a temperature in the internal space of the tent **200** may be adjusted.

(16) That is, the duct **300** may be provided in a form of a hose. The duct **300** extends so that the interior of the mobility vehicle **100** is connected to the internal space of the tent **200** and the air-conditioned air provided through the air conditioner **110** flows into the internal space of the tent **200**.

(17) Therefore, a user controls the air conditioner **110** of the mobility vehicle **100** to provide the cooling air or the heating air at the time of adjusting the temperature in the internal space of the tent **200** so that the air-conditioned air may flow to the tent **200** through the duct **300** and the temperature in the internal space of the tent **200** may be adjusted.

(18) In detail, the mobility vehicle **100** has an air intake port through which internal air is introduced, and an air discharge port through which the air-conditioned air provided by the air conditioner **110** is discharged.

(19) Furthermore, the duct **300** includes an inlet duct **310** and an outlet duct **320**. The inlet duct **310** is connected to the air discharge port to allow the air-conditioned air to flow to the tent **200**, and the outlet duct **320** is connected to the air intake port to allow the air in the tent **200** to circulate to the mobility vehicle **100**.

(20) As described above, the duct **300** includes the inlet duct **310** configured to allow the air-conditioned air in the mobility vehicle **100** to flow to the duct **300**, and the outlet duct **320** configured to allow the air-conditioned air to circulate back to the mobility vehicle **100** after adjusting the temperature in the tent **200**.

(21) Therefore, the air-conditioned air provided by the air conditioner **110** of the mobility vehicle **100** flows to the tent **200** through the air discharge port and the inlet duct **310**, adjusts the temperature in the internal space of the tent **200**, and then circulates to the air intake port of the mobility vehicle **100** through the outlet duct **320**. Therefore, the comfort of the internal space of the tent **200** is improved as the air-conditioned air circulates through the tent **200**.

(22) Meanwhile, as illustrated in FIG. **2**, one end portion of the inlet duct **310** is detachably connected to the air discharge port of the mobility vehicle **100**, and the other end portion of the inlet duct **310** is penetratively connected to the tent **200**.
(23) For example, a hook structure may be applied to one end portion of the inlet duct **310** so that one end portion of the inlet duct **310** may be mounted on the air discharge port by hook connection. The inlet duct **310** may be detachably mounted on the air discharge port by various methods such as a magnet and a Velcro fastener in addition to the hook structure. Furthermore, the other end portion of the inlet duct **310** may be penetratively connected to the tent **200**. A fitting coupling structure or a catching structure may be applied to the other end portion of the inlet duct **310** so that the other end portion of the inlet duct **310** may be fixedly mounted in the tent **200**.
(24) In the instant case, the inlet duct **310** is formed to surround the air discharge port and mounted to cover the air discharge port so that the air-conditioned air may be provided to the inlet duct **310** in a state in which a loss of the air-conditioned air discharged through the air discharge port is minimized. That is, the inlet duct **310** may be larger in size than the air discharge port and cover the air discharge port so that the air-conditioned air discharged through the air discharge port may flow to the inlet duct **310** in the state in which a loss of the air-conditioned air is minimized.
(25) Meanwhile, in a case in which the air-conditioned air is provided through a front air conditioner **110***a*, the inlet duct **310** is coupled to a defrosting discharge port among air discharge ports. Of course, the inlet duct **310** may be mounted on various air discharge ports provided in the mobility vehicle **100**. However, the inlet duct for supplying the air-conditioned air supplied from the front air conditioner **110***a* is mounted on the defrosting discharge port to ensure a flow rate of the air-conditioned air and make it easy to mount the inlet duct.
(26) Furthermore, when the inlet duct **310** is provided on the air discharge port corresponding to an occupant in a state in which the occupant is present in the mobility vehicle **100**, air conditioning efficiency for the occupant may deteriorate. Therefore, the inlet duct **310** is mounted on the defrosting discharge port so that the air conditioning efficiency is also ensured even in the interior of the mobility vehicle **100**.
(27) Furthermore, one end portion of the outlet duct **320** is also penetratively connected to the tent **200**. A fitting coupling structure or a catching structure may be applied to one end portion of the outlet duct **320** so that the state in which one end portion of the outlet duct **320** is mounted to the tent **200** may be fixed. Furthermore, the air intake port and the other end portion of the outlet duct **320** may be detachably connected by various methods such as a hook connection structure, a magnet, and a Velcro fastener.
(28) Meanwhile, as illustrated in FIG. **3**, in a case in which the air-conditioned air is provided through a rear air conditioner **110***b*, the inlet duct **310** may be connected to an air discharge port connected to the rear air conditioner **110***b*, and the outlet duct **320** may be connected to an air inlet port connected to the rear air conditioner **110***b*.
(29) As described above, to use the air-conditioned air supplied from the rear air conditioner **110***b*, the inlet duct **310** and the outlet duct **320** are respectively connected to the air discharge port and the air inlet port provided in a rear seat so that the air-conditioned air supplied from the rear air conditioner **110***b* may be transmitted to the tent **200**.
(30) Therefore, in the case in which the occupant is present in the interior of the mobility vehicle **100**, the air-conditioned air supplied from the front air conditioner **110***a* may be supplied to the interior of the mobility vehicle **100**, and the air-conditioned air supplied from the rear air conditioner **110***b* may be supplied to the tent **200** through the inlet duct **310** and the outlet duct **320**.
(31) As described above, according to an exemplary embodiment of the present disclosure, to provide the air-conditioned air to the tent **200**, the air-conditioned air supplied from the front air conditioner **110***a* or the rear air conditioner **110***b* may be transmitted to the tent **200** through the inlet duct **310** and the outlet duct **320**. Furthermore, the inlet duct **310** and the outlet duct **320** are selectively and respectively connected to the air discharge port and the air intake port depending on

whether the front air conditioner **110***a* or the rear air conditioner **110***b* is used according to the situations, ensuring the utilization of the air conditioning air.

(32) Meanwhile, the apparatus further includes a duct bracket **400** detachably mounted on the mobility vehicle **100** and configured to fix a position of the duct **300**.

(33) As illustrated in FIG. **4**, the duct bracket **400** is configured so that the inlet duct **310** and the outlet duct **320**, which form the duct **300**, penetrate the duct bracket **400**. The duct bracket **400** fixes the positions of the inlet duct **310** and the outlet duct **320** by surrounding the inlet duct **310** and the outlet duct **320**.

(34) That is, the inlet duct **310** and the outlet duct **320**, which form the duct **300**, pass through an opening/closing unit **120** including a door glass or a roof of the mobility vehicle **100** to allow the interior of the mobility vehicle **100** and the internal space of the tent **200** to fluidically communicate with each other.

(35) In the instant case, the opening/closing unit **120** may be a door glass or sunroof. That is, the inlet duct **310** and the outlet duct **320** are connected to the interior of the mobility vehicle **100** from the exterior of the mobility vehicle **100**. Therefore, the inlet duct **310** and the outlet duct **320** pass through the door glass or sunroof including a relatively small open region except for a door so that the air-conditioned air flows between the interior of the mobility vehicle **100** and the internal space of the tent **200**.

(36) Of course, the inlet duct **310** and the outlet duct **320** may pass through the internal and the external through the opened door. However, air conditioning efficiency deteriorates because of heat exchange between the internal and the external. Therefore, the inlet duct **310** and the outlet duct **320** connect the mobility vehicle **100** and the tent **200** through the door glass or sunroof.

(37) Therefore, the duct bracket **400** is mounted on the opening/closing unit **120** including the door glass or the roof of the mobility vehicle **100**, and the duct **300** passes through the duct bracket **400** to allow the interior of the mobility vehicle **100** and the internal space of the tent **200** to fluidically communicate with each other.

(38) That is, the duct bracket **400** is mounted on the opening/closing unit **120**, and the inlet duct **310** and the outlet duct **320**, which form the duct **300**, penetrate the duct bracket **400** so that the remaining region, except for the region of the opening/closing unit **120** penetrated by the inlet duct **310** and the outlet duct **320**, is closed. Therefore, the interior of the mobility vehicle **100** is sealed, and the air conditioning efficiency in the internal is ensured.

(39) The duct bracket **400** may have holes through which the inlet duct **310** and the outlet duct **320** pass. A sealing body for sealing a portion between the inlet duct **310** and the outlet duct **320** may be further provided at the periphery of the holes.

(40) Furthermore, the duct bracket **400** is detachably mounted on the opening/closing unit **120** so that the duct bracket **400** may be selectively used only in the camping situation in which the air-conditioned air flows between the mobility vehicle **100** and the tent **200**. The positions of the inlet duct **310** and the outlet duct **320** are fixed by the duct bracket **400**, which prevents damage to the components due to free movements of the inlet duct **310** and the outlet duct **320**.

(41) Furthermore, the duct bracket **400** may be formed to match a shape of a portion of the door glass or the roof forming the opening/closing unit **120** and mounted by being pressed against the door glass or the roof when the opening/closing unit **120** is closed.

(42) That is, because the duct bracket **400** is formed to match a portion of the overall shape of the door glass or the roof forming the opening/closing unit **120**, the duct bracket **400** blocks the remaining region even though the opening/closing unit **120** is not completely closed when the opening/closing unit **120** is closed.

(43) Furthermore, because the state in which the duct bracket **400** is mounted by being pressed against the door glass or the roof forming the opening/closing unit **120** is maintained at the time of closing the opening/closing unit **120**, a separate means for fixing the duct bracket **400** to the opening/closing unit **120** is not required. Furthermore, the duct bracket **400** is formed so that the

door glass or the roof forming the opening/closing unit **120** is inserted into a rim portion of the duct bracket **400** so that the duct bracket **400** may be securely mounted on the opening/closing unit **120**.
(44) As described above, the apparatus of sharing air-conditioned air for the mobility vehicle **100** according to an exemplary embodiment of the present disclosure allows the mobility vehicle **100** and the tent **200** to share the air-conditioned air through the duct **300** so that the air-conditioned air produced in the mobility vehicle **100** circulates through the duct **300** and the temperature in the internal space of the tent **200** is adjusted. Furthermore, energy efficiency is improved as a loss of the air-conditioned air is minimized at the time of transmitting the air-conditioned air from the mobility vehicle **100** to the tent **200**.
(45) Meanwhile, as illustrated in FIGS. **1** and **5**, a system for sharing air-conditioned air for the mobility vehicle **100** according to various exemplary embodiments of the present disclosure may include the mobility vehicle **100** including the air conditioner **110** configured to provide the air-conditioned air to the interior of the mobility vehicle **100**, the mobility vehicle **100** including the air intake port through which the internal air is introduced, and the air discharge port through which the air-conditioned air provided by the air conditioner **110** is discharged; the tent **200** provided outside the interior of the mobility vehicle **100** and including the internal space; and the duct **300** extending to allow the interior of the mobility vehicle **100** and the internal space of the tent **200** to fluidically communicate with each other and configured to allow the air-conditioned air provided by the air conditioner **110** to be shared with the internal space of the tent **200**. The system further includes a control unit **500** configured to control the mobility vehicle **100** including the air conditioner **110**, include a camping mode in addition to an air conditioning mode, and control the air conditioner **110** according to the camping mode when the camping mode is selected.
(46) That is, according to an exemplary embodiment of the present disclosure, when the user selects the camping mode in the state in which the mobility vehicle **100** and the tent **200** are connected to fluidically communicate with each other through the duct **300**, the air-conditioned air provided by the air conditioner **110** flows to the tent **200** through the duct **300** so that the temperature in the internal space of the tent **200** is adjusted.
(47) In the instant case, when the user selects the camping mode through a mobile phone terminal held by the user or a manipulation unit provided in the mobility vehicle **100**, the control unit **500** controls the air conditioner **110** according to the camping mode.
(48) The control according to the selected camping mode may be performed as follows.
(49) When the camping mode is selected, the control unit **500** performs a recirculation mode in which outside air is blocked and internal air is circulated. That is, the control unit **500** performs the recirculation mode when the camping mode is selected so that only the internal air circulates in the interior of the mobility vehicle **100**, which improves cooling and heating efficiency.
(50) Furthermore, when the camping mode is selected, the control unit **500** allows the air-conditioned air to flow at a maximum flow rate. That is, when the camping mode is selected, the air-conditioned air discharged from the air discharge port needs to move to the tent **200** through the duct **300**. Therefore, a blower is maximally operated to allow the air-conditioned air to smoothly flow to the tent **200** through the duct **300**.
(51) Meanwhile, the duct **300** includes the inlet duct **310** and the outlet duct **320**. The inlet duct **310** is connected to the air discharge port to allow the air-conditioned air to flow to the tent **200**, and the outlet duct **320** is connected to the air intake port to allow the air in the tent **200** to circulate to the mobility vehicle **100**.
(52) Therefore, the air-conditioned air provided by the air conditioner **110** of the mobility vehicle **100** flows to the tent **200** through the air discharge port and the inlet duct **310**, adjusts the temperature in the internal space of the tent **200**, and then circulates to the air intake port of the mobility vehicle **100** through the outlet duct **320**. Therefore, the comfort of the internal space of the tent **200** is improved as the air-conditioned air circulates through the tent **200**.
(53) In the instant case, when the camping mode is selected, the control unit **500** may perform

control to open the air discharge port connected to the inlet duct **310** and close the remaining air discharge ports.
(54) That is, when the camping mode is selected, the air-conditioned air provided by the air conditioner **110** of the mobility vehicle **100** flows to the tent **200** through the inlet duct **310**. Therefore, the control unit **500** controls respective doors in the air conditioner **110** to open only the air discharge port connected to the inlet duct **310** and close the remaining air discharge ports.
(55) Therefore, the air-conditioned air is concentrated only in the air discharge port connected to the inlet duct **310** and then discharged so that a flow rate of the air-conditioned air flowing to the tent **200** is ensured, which improves performance in conditioning air in the tent **200**.
(56) Meanwhile, the control unit **500** receives information as to whether the occupant is present in the interior of the mobility vehicle **100**. When the occupant is present in the interior of the mobility vehicle **100**, the control unit **500** performs control to open the air discharge port corresponding to a seat in which the occupant is accommodated.
(57) That is, the control unit **500** may determine whether the occupant is present in the interior of the mobility vehicle **100** based on information obtained from a pressure detector provided in a seat, information obtained from a camera detector provided in the mobility vehicle **100**, or information as to whether a seat is folded.
(58) Therefore, when the occupant is present in the interior of the mobility vehicle, the control unit **500** performs control to open the air discharge port corresponding to the seat in which the occupant is accommodated so that the air-conditioned air is also provided to the occupant in the interior of the mobility vehicle.
(59) Therefore, the air discharge port, which is connected to the inlet duct **310**, and the air discharge port, which corresponds to the seat in which the occupant is accommodated, are opened when the occupant is present in the internal in the state in which the camping mode is selected so that the air-conditioned air flows to the tent **200** and the occupant in, the interior, allowing the air-conditioned air to ensure the comfort in each of the spaces.
(60) Meanwhile, when the camping mode is selected, the control unit **500** checks, from the user, whether the inlet duct **310** and the outlet duct **320** are mounted. When it is determined, from the user, that the inlet duct **310** and the outlet duct **320** are mounted, the control unit **500** performs control according to the camping mode.
(61) That is, when the camping mode is selected, the control unit **500** requests the user to mount the inlet duct **310** and the outlet duct **320** through the user's mobile phone terminal or a display in the mobility vehicle **100**. Thereafter, when the user inputs a command in respect to the completion of the mounting of the duct **300** through the mobile phone terminal or the display in the state in which the inlet duct **310** is mounted on the air discharge port and the outlet duct **320** is mounted on the air intake port, the control unit **500** determines that the inlet duct **310** and the outlet duct **300** are normally mounted, and the control unit **500** performs control according to the camping mode.
(62) As described above, when the camping mode is selected, the control unit **500** completes a preparation process according to the camping mode and then controls the air conditioner **110** according to the camping mode, preventing a malfunction.
(63) Meanwhile, the mobility vehicle **100** further includes the duct bracket **400** detachably mounted on the opening/closing unit **120** including the door glass or roof. The inlet duct **310** and the outlet duct **320** penetrate the duct bracket **400**, and the duct bracket **400** fixes the inlet duct **310** and the outlet duct **320**.
(64) That is, the duct bracket **400** is mounted on the opening/closing unit **120**, and the inlet duct **310** and the outlet duct **320** penetrate the duct bracket **400** so that the remaining region, except for the region of the opening/closing unit **120** penetrated by the inlet duct **310** and the outlet duct **320**, is closed.
(65) Furthermore, the duct bracket **400** is detachably mounted on the opening/closing unit **120** so that the duct bracket **400** may be selectively used only in the camping situation in which the air-

conditioned air flows between the mobility vehicle **100** and the tent **200**. The positions of the inlet duct **310** and the outlet duct **320** are fixed by the duct bracket **400**, which prevents damage to the components due to free movements of the inlet duct **310** and the outlet duct **320**.
(66) Therefore, when the camping mode is selected, the control unit **500** checks whether the duct bracket **400** is mounted on the opening/closing unit **120**. When the duct bracket **400** is mounted on the opening/closing unit **120**, the control unit **500** performs control according to the camping mode.
(67) In detail, when an opening amount of the opening/closing unit **120** is at a predetermined level, the control unit **500** determines that the duct bracket **400** is mounted, and the control unit **500** performs control according to the camping mode when it is determined that the duct bracket **400** is mounted.
(68) That is, the duct bracket **400** may be formed to match a shape of a portion of the door glass or the roof forming the opening/closing unit **120** and mounted by being pressed against the door glass or the roof when the opening/closing unit **120** is closed. Therefore, the control unit **500** may check whether the door glass or the roof forming the opening/closing unit **120** is closed at a predetermined level or more to fix the duct bracket **400**, determining whether the duct bracket **400** is mounted on the opening/closing unit **120**.
(69) Furthermore, when the user inputs the command in respect to the completion of the mounting of the duct bracket **400** through the mobile phone terminal or the display in the state in which the duct bracket **400** is mounted on the opening/closing unit **120**, the control unit **500** determines that the duct bracket **400** is normally mounted, and the control unit **500** performs control according to the camping mode.
(70) Meanwhile, when the camping mode is selected, the control unit **500** performs control not to perform a function of removing moisture. That is, the mobility vehicle **100** is provided with a function of removing fog such as an Auto defog system (ADS). However, because the user is in the tent **200** in the camping mode, there is no problem even though the interior of the mobility vehicle **100** fogs up.
(71) An unnecessary loss of electric power occurs when the function of removing moisture is performed in the camping mode as described above, the control unit **500** does not operate the ADS when the camping mode is selected.
(72) Meanwhile, when the camping mode is selected, the control unit **500** receives information in a state of charge (SOC) value of the battery of the mobility vehicle **100** and checks a charging station closest to the current position of the mobility vehicle **100** or the minimum amount of electricity of the battery which is to be consumed while the mobility vehicle **100** gets a preset charging station. When the state of charge of the battery reaches the minimum amount of electricity of the battery, the control unit **500** does not operate the air conditioner **110**.
(73) In the instant case, the control unit **500** checks information on the state of charge (SOC) value of the battery and checks the amount of consumed electricity of the battery at the time of controlling the air conditioner **110** according to the camping mode.
(74) Furthermore, the control unit **500** checks the charging station closest to the current position of the mobility vehicle **100** or the minimum amount of electricity of the battery which is to be consumed when the mobility vehicle **100** gets to the preset charging station or a charging station lastly used by the mobility vehicle **100**.
(75) Therefore, the control unit **500** prevents the discharge of the battery caused by the consumption of the electric power when the air conditioner **110** operates in the camping mode so that the state of charge of the battery, which enables the mobility vehicle **100** to move to the charging station, is ensured, and the use of the mobility vehicle **100** is stabilized.
(76) Furthermore, when the state of charge of the battery reaches the minimum amount of electricity of the battery, the control unit **500** does not operate the air conditioner **110** and stops operations of other devices, which consume electric power, in addition to the air conditioner **110**, facilitating the mobility vehicle **100** to stably move to the charging station and be charged.

(77) Therefore, the control unit **500** may implement the camping mode according to control steps S**1**, S**2**, S**3**, S**4**, S**5**, S**6**, S**7**, S**8**, S**9**, S**10**, S**11**, S**12** and S**13** in the flowchart illustrated in FIG. **6**.
(78) The apparatus and system for sharing air-conditioned air for the mobility vehicle **100** structured as described above allow the mobility vehicle **100** and the tent **200** to share the air-conditioned air through the duct **300** so that the air-conditioned air produced in the mobility vehicle **100** circulates through the duct **300** and the temperature in the internal space of the tent **200** is adjusted.
(79) Furthermore, energy efficiency is improved as a loss of the air-conditioned air is minimized at the time of transmitting the air-conditioned air from the mobility vehicle **100** to the tent **200**.
(80) Furthermore, the term related to a control device such as “controller”, “control apparatus”, “control unit”, “control device”, “control module”, or “server”, etc refers to a hardware device including a memory and a processor configured to execute one or more steps interpreted as an algorithm structure. The memory stores algorithm steps, and the processor executes the algorithm steps to perform one or more processes of a method in accordance with various exemplary embodiments of the present disclosure. The control device according to exemplary embodiments of the present disclosure may be implemented through a nonvolatile memory configured to store algorithms for controlling operation of various components of a vehicle or data about software commands for executing the algorithms, and a processor configured to perform operation to be described above using the data stored in the memory. The memory and the processor may be individual chips. Alternatively, the memory and the processor may be integrated in a single chip. The processor may be implemented as one or more processors. The processor may include various logic circuits and operation circuits, may process data according to a program provided from the memory, and may generate a control signal according to the processing result.
(81) The control device may be at least one microprocessor operated by a predetermined program which may include a series of commands for carrying out the method included in the aforementioned various exemplary embodiments of the present disclosure.
(82) The aforementioned invention can also be embodied as computer readable codes on a computer readable recording medium. The computer readable recording medium is any data storage device that can store data which may be thereafter read by a computer system and store and execute program instructions which may be thereafter read by a computer system. Examples of the computer readable recording medium include Hard Disk Drive (HDD), solid state disk (SSD), silicon disk drive (SDD), read-only memory (ROM), random-access memory (RAM), CD-ROMs, magnetic tapes, floppy discs, optical data storage devices, etc and implementation as carrier waves (e.g., transmission over the Internet). Examples of the program instruction include machine language code such as those generated by a compiler, as well as high-level language code which may be executed by a computer using an interpreter or the like.
(83) In various exemplary embodiments of the present disclosure, each operation described above may be performed by a control device, and the control device may be configured by multiple control devices, or an integrated single control device.
(84) In various exemplary embodiments of the present disclosure, the control device may be implemented in a form of hardware or software, or may be implemented in a combination of hardware and software.
(85) Furthermore, the terms such as “unit”, “module”, etc. Included in the specification mean units for processing at least one function or operation, which may be implemented by hardware, software, or a combination thereof.
(86) For convenience in explanation and accurate definition in the appended claims, the terms “upper”, “lower”, “inner”, “outer”, “up”, “down”, “upwards”, “downwards”, “front”, “rear”, “back”, “inside”, “outside”, “inwardly”, “outwardly”, “interior”, “exterior”, “internal”, “external”, “forwards”, and “backwards” are used to describe features of the exemplary embodiments with reference to the positions of such features as displayed in the figures. It will be further understood

that the term “connect” or its derivatives refer both to direct and indirect connection.
(87) The foregoing descriptions of specific exemplary embodiments of the present disclosure have been presented for purposes of illustration and description. They are not intended to be exhaustive or to limit the present disclosure to the precise forms disclosed, and obviously many modifications and variations are possible in light of the above teachings. The exemplary embodiments were chosen and described to explain certain principles of the present disclosure and their practical application, to enable others skilled in the art to make and utilize various exemplary embodiments of the present disclosure, as well as various alternatives and modifications thereof. It is intended that the scope of the present disclosure be defined by the Claims appended hereto and their equivalents.

Claims

1. An apparatus comprising: a mobility vehicle including an air conditioner configured to provide air-conditioned air, the apparatus further comprising: the mobility vehicle including the air conditioner configured to provide the air-conditioned air to an interior of the mobility vehicle; a shelter provided outside the interior of the mobility vehicle and including an internal space in the shelter; and at least one duct configured for extending to allow the interior of the mobility vehicle and the internal space of the shelter to fluidically communicate with each other, the at least one duct being configured to allow the air-conditioned air provided by the air conditioner to be shared with the internal space of the shelter.
2. The apparatus of claim 1, wherein the air conditioner includes: an air intake port through which internal air of the mobility vehicle is introduced; and at least one air discharge port through which the air-conditioned air provided by the air conditioner is discharged, wherein the at least one duct comprises an inlet duct and an outlet duct, and wherein the inlet duct is connected to the at least one air discharge port to allow the air-conditioned air to flow to the shelter, and the outlet duct is connected to the air intake port to allow air in the shelter to circulate through the mobility vehicle.
3. The apparatus of claim 2, wherein a first end portion of the inlet duct is detachably connected to the at least one air discharge port of the mobility vehicle, and a second end portion of the inlet duct is penetratively connected to the shelter, and wherein a first end portion of the outlet duct is penetratively connected to the shelter, and a second end portion of the outlet duct is detachably connected to the air intake port of the mobility vehicle.
4. The apparatus of claim 2, wherein the inlet duct is formed to surround the at least one air discharge port and mounted to cover the at least one air discharge port.
5. The apparatus of claim 2, wherein the at least one air discharge port includes a plurality of air discharge ports, and wherein the air conditioner includes a front air conditioner, and wherein when the air-conditioned air is provided through the front air conditioner, the inlet duct is coupled to a defrosting discharge port among the plurality of air discharge ports.
6. The apparatus of claim 2, wherein the air conditioner includes a rear air conditioner, and wherein when the air-conditioned air is provided through the rear air conditioner, the inlet duct is connected to the at least one air discharge port connected to the rear air conditioner, and the outlet duct is connected to the air inlet port connected to the rear air conditioner.
7. The apparatus of claim 1, further including: a duct bracket detachably mounted on the mobility vehicle and configured to fix a position of the at least one duct.
8. The apparatus of claim 7, wherein the duct bracket is detachably mounted on an opening/closing unit including a door glass or a roof of the mobility vehicle, and the at least one duct passes through the duct bracket to allow the interior of the mobility vehicle and the internal space of the shelter to fluidically communicate with each other through the at least one duct.
9. The apparatus of claim 8, wherein the duct bracket is formed to match a shape of a portion of the door glass or the roof forming the opening/closing unit, and the duct bracket is mounted by being

pressed against the door glass or the roof when the opening/closing unit is closed.
10. A system comprising: a mobility vehicle including an air conditioner configured to provide air-conditioned air, the system further including: the mobility vehicle including the air conditioner configured to provide the air-conditioned air to an interior of the mobility vehicle, the air conditioner including an air intake port through which internal air of the mobility vehicle is introduced, and an air discharge port through which the air-conditioned air provided by the air conditioner is discharged; a shelter provided outside the interior of the mobility vehicle and including an internal space in the shelter; at least one duct configured for extending to allow the interior of the mobility vehicle and the internal space of the shelter to fluidically communicate with each other, the at least one duct being configured to allow the air-conditioned air provided by the air conditioner to be shared with the internal space of the shelter; and a control unit configured to control the mobility vehicle including the air conditioner, under a camping mode or an air conditioning mode, and to control the air conditioner according to the camping mode when the camping mode is selected.
11. The system of claim 10, wherein when the camping mode is selected, the control unit is configured to perform a recirculation mode in which outside air is blocked out of the mobility vehicle and the internal air circulates in the mobility vehicle.
12. The system of claim 10, wherein when the camping mode is selected, the control unit is configured to control the air conditioner to blow the air-conditioned air at a maximum flow rate.
13. The system of claim 10, wherein the at least one duct comprises an inlet duct and an outlet duct, the inlet duct is connected to the air discharge port to allow the air-conditioned air to flow to the shelter, and the outlet duct is connected to the air intake port to allow air in the shelter to circulate through the mobility vehicle, and wherein when the camping mode is selected, the control unit is configured to perform control of the air conditioner to open the air discharge port connected to the inlet duct and close a remaining air discharge port of the air conditioner.
14. The system of claim 13, wherein the control unit is configured to receive information as to whether an occupant is present in the interior of the mobility vehicle, and wherein when the control unit concludes that the occupant is present in the interior of the mobility vehicle, the control unit is configured to perform control of the air conditioner to open the air discharge port corresponding to a seat in which the occupant is accommodated.
15. The system of claim 13, wherein when the camping mode is selected, the control unit is configured to check whether the inlet duct and the outlet duct are mounted, and wherein when the control unit concludes that the inlet duct and the outlet duct are mounted, the control unit is configured to perform control of the air conditioner according to the camping mode.
16. The system of claim 15, wherein, when the control unit receives an input from a user related to completion of the mounting the inlet duct on the air discharge port and the outlet duct on the air intake port, the control unit concludes that the inlet duct and the outlet duct are mounted.
17. The system of claim 13, wherein the mobility vehicle further includes a duct bracket detachably mounted on an opening/closing unit including a door glass or a roof of the mobility vehicle, the duct bracket is configured to fix the inlet duct and the outlet duct as the inlet duct and the outlet duct penetrate the duct bracket, and the control unit is configured to check whether the duct bracket is mounted on the opening/closing unit when the camping mode is selected.
18. The system of claim 17, wherein when an opening amount of the opening/closing unit is at a predetermined level, the control unit is configured to determine that the duct bracket is mounted, and wherein when the control unit determines that the duct bracket is mounted, the control unit is configured to perform control of the air conditioner according to the camping mode.
19. The system of claim 10, wherein when the camping mode is selected, the control unit is configured to perform control of the air conditioner not to perform a function of removing moisture.
20. The system of claim 10, wherein when the camping mode is selected, the control unit is

configured to receive information in a state of charge (SOC) value of a battery of the mobility vehicle and configured to check a charging station closest to a current position of the mobility vehicle or a minimum amount of electricity of the battery which is to be consumed while the mobility vehicle gets to a preset charging station, and wherein when the SOC value of the battery reaches the minimum amount of electricity of the battery, the control unit is configured not to operate the air conditioner.